

Binder Generation with FreeBindCraft

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1. Prerequisites

1.1 Install VS Code

Download and install VS Code from the official download page:

https://code.visualstudio.com/download?_exp_download=fb315fc982

1.2 Github Account

Create a GitHub account at <https://github.com/> if you do not already have one.

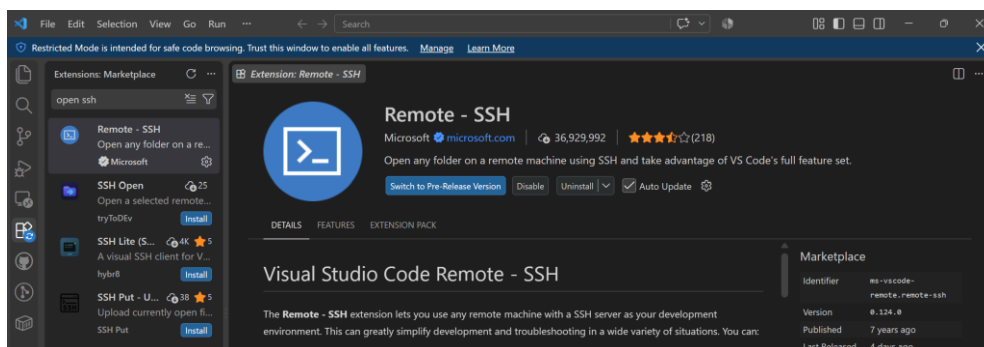
1.3 La Trobe VPN

To securely connect to Ewok machine off campus, download and setup the VPN here:

<https://www.latrobe.edu.au/students/support/it/connecting/vpn>

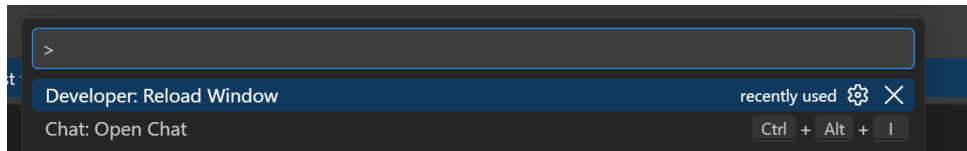
1.4 Remote- SSH extension

Install the **Remote – SSH** extension in VS Code. This extension enables you to connect to an Ewok machine from your local computer and access its computational resources directly through VS Code.



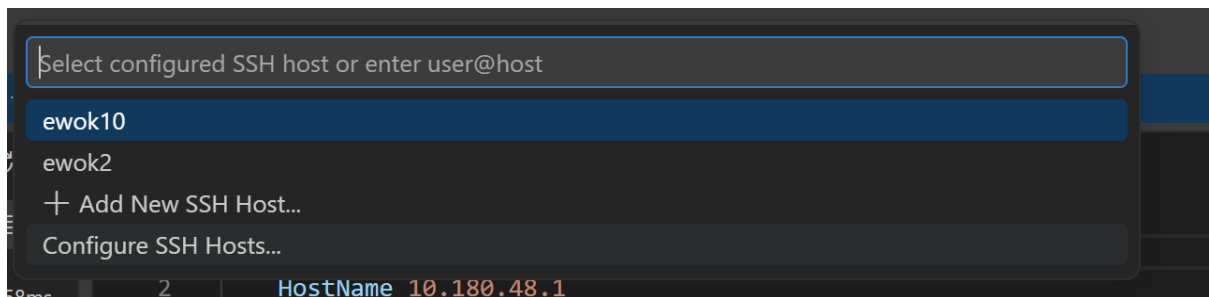
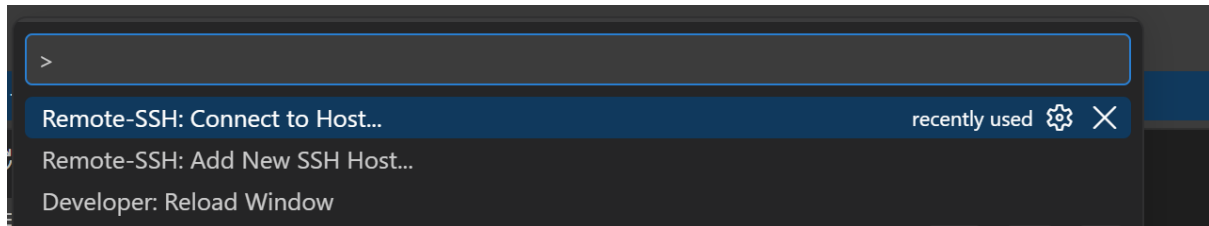
After installation, open the **Command Palette** (**Ctrl + Shift + P** on Windows or **Cmd + Shift + P** on Mac.) and type **>Developer: Reload Window**, and press Enter to ensure the extension loads correctly.

Quick tip: To run any command in the **Command Palette**, make sure to start it with a **>** symbol, e.g. **>your command**.



2. Connect to the Ewok Machine

In VS Code, open the **Command Palette** and type **>Remote-SSH: Connect to Host**, then select **Configure SSH Hosts**.



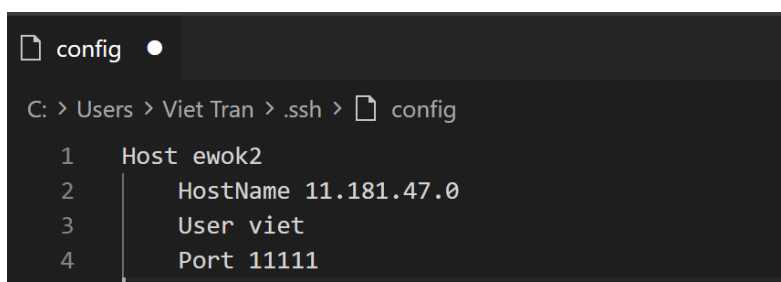
Add an entry for your Ewok machine using the following configuration format:

```
Host <ewok_machine_number>
  HostName <your_ewok_machine_ip_address>
  User <your_ewok_username>
  Port <your_ewok_port>
```

Example configuration:

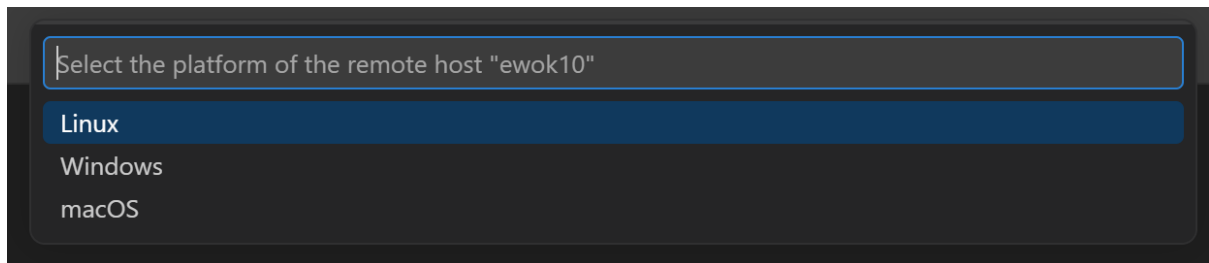
```
Host ewok2
  HostName 11.181.47.0
  User viet
  Port 11111
```

Make sure to save your changes to the config via **Ctrl + S** on Windows or **Cmd + S** on Mac.

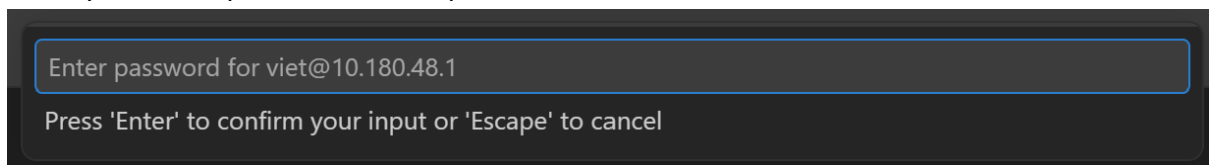


Note: Contact [Zhen He](#) or [Aaron Harris](#) for setting up and access your Ewok account.

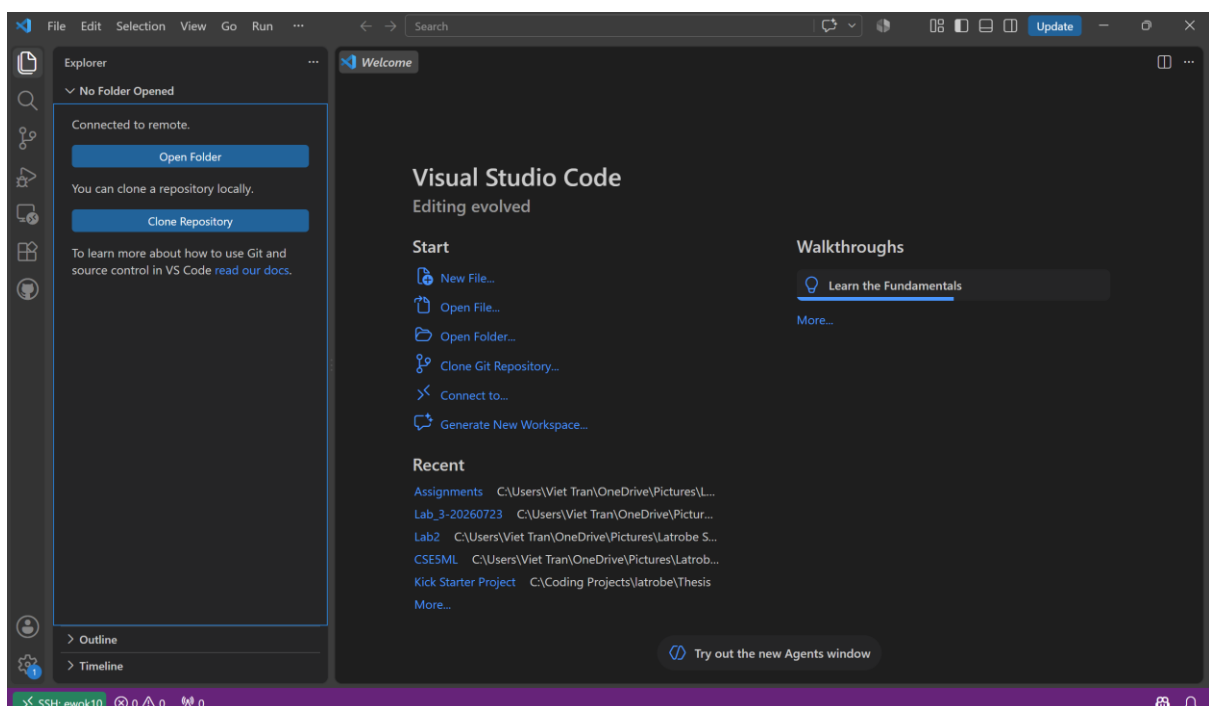
If VS Code prompts you to select the remote host's operating system, choose **Linux**



Enter your Ewok password to complete the connection.



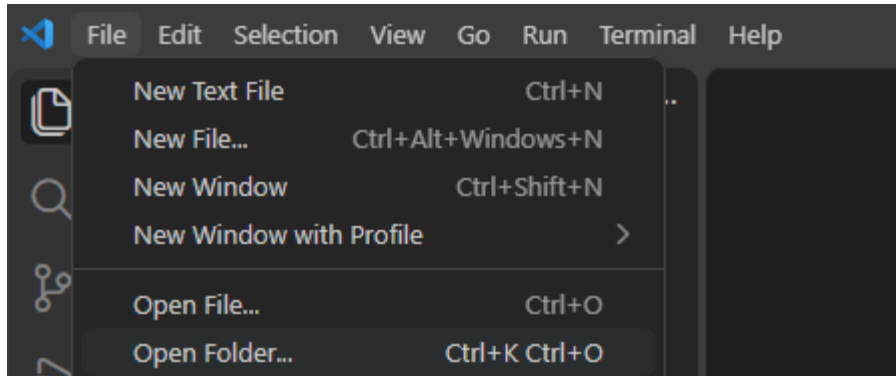
Once the connection is established, the Ewok host name should appear in the bottom-left corner of the VS Code window.



3. Binders Generation with FreeBindCraft

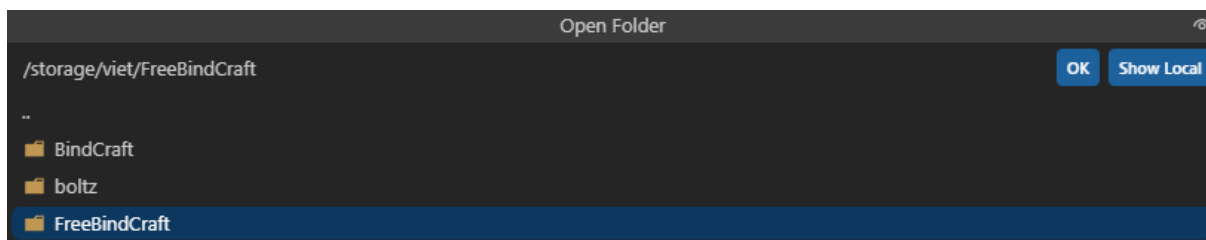
3.1 Open the FreeBindCraft Project

After connecting to the Ewok machine, select **File** → **Open Folder** in VS Code.



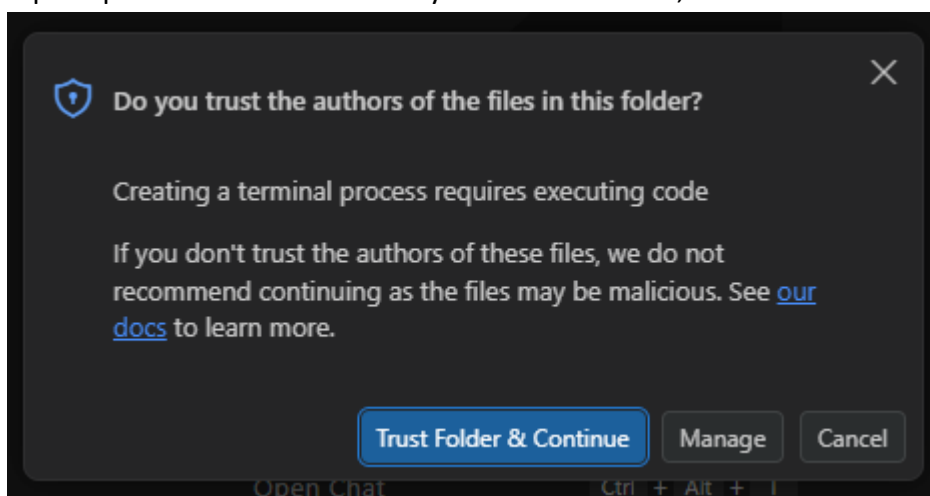
Open the directory in which FreeBindCraft is installed. Typical installation locations include:

- /home/<username>/FreeBindCraft
- /storage/<username>/FreeBindCraft



Replace <username> with your Ewok username.

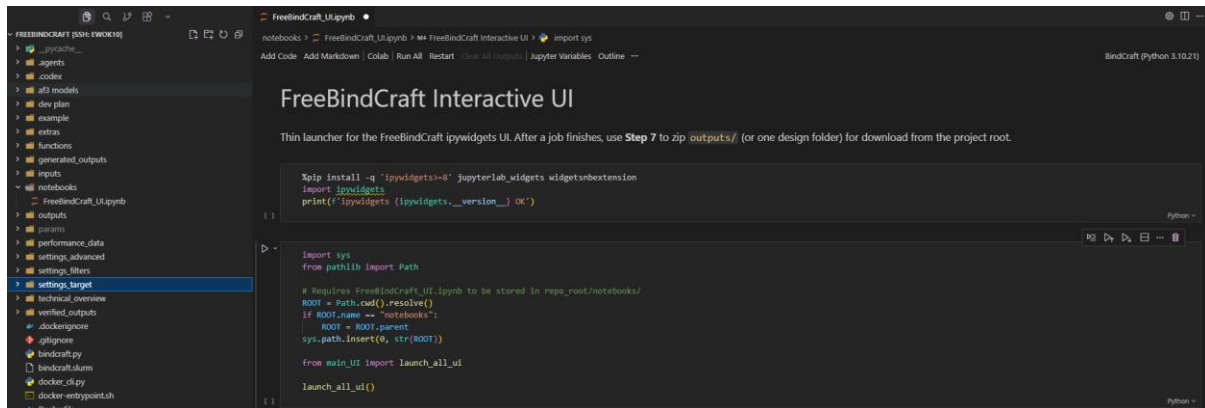
If prompted to confirm whether you trust the folder, select **Trust Folder & Continue**.



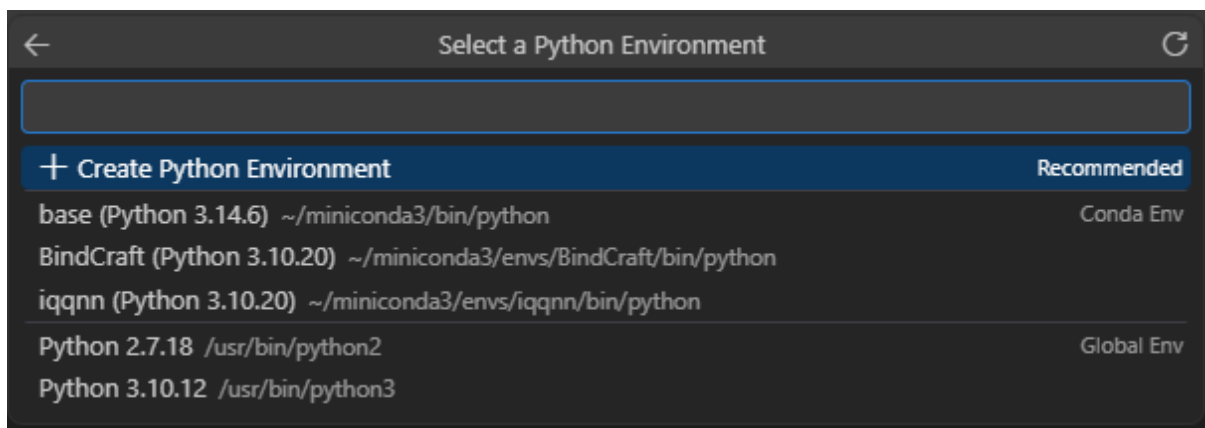
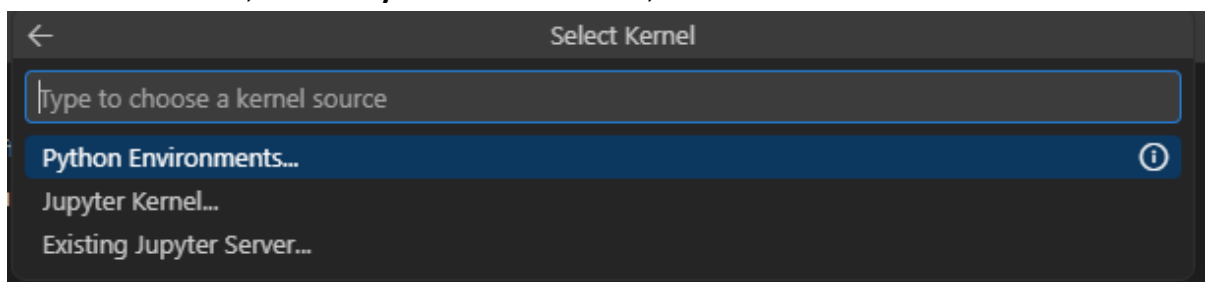
3.2 FreeBindCraft UI

3.2.1 Set Up the Notebook

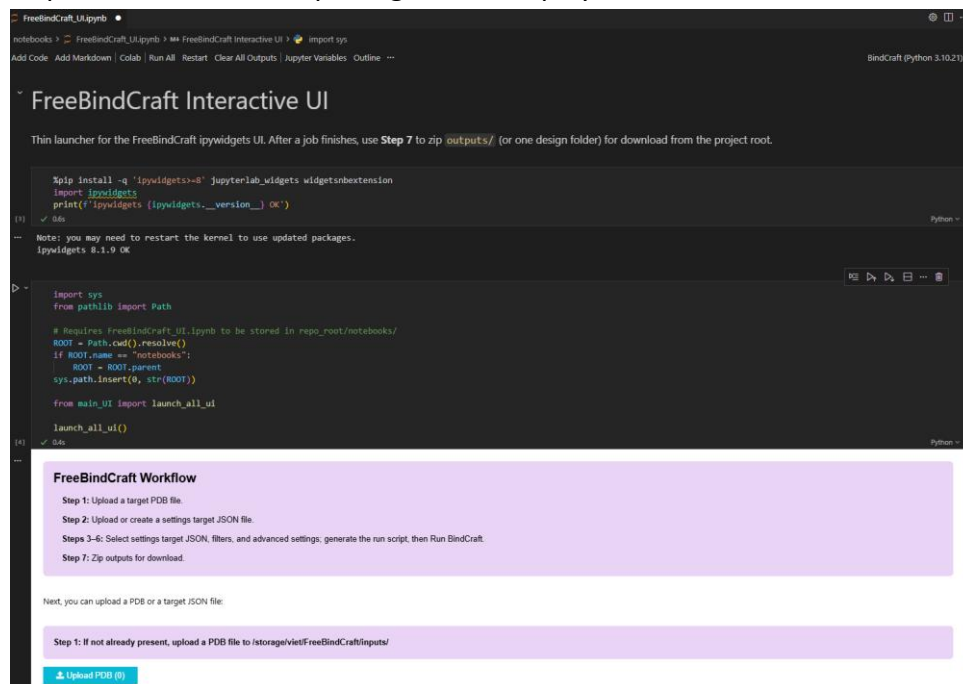
In the **Explorer** panel on the left side of VS Code, open notebooks/FreeBindCraft_UI.ipynb



Click **Select Kernel**, choose **Python Environments**, and select the **BindCraft** environment.



After kernel successfully connected, click **Run All** to execute the notebook cells, install the required user interface packages, and display the interactive FreeBindCraft interface.



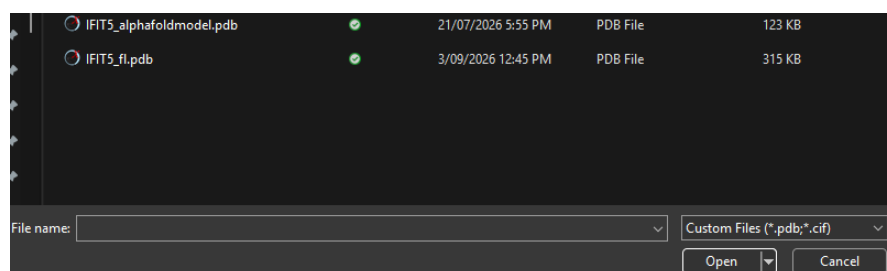
3.2.2 Run a FreeBindCraft job via the UI

Step 1: Upload a Target PDB file

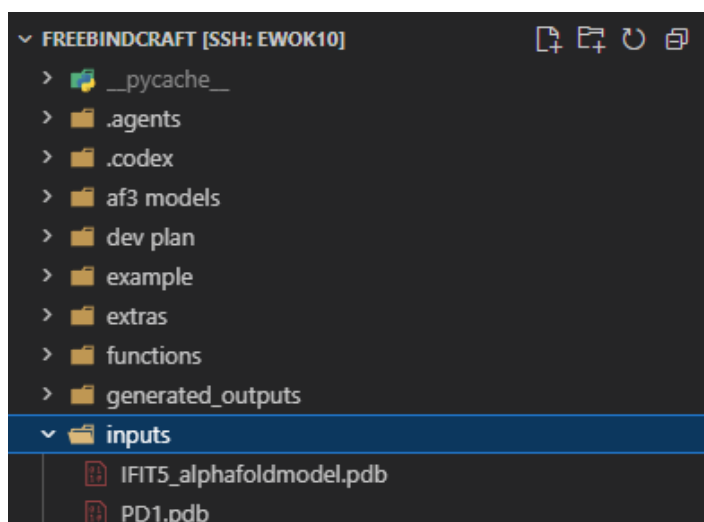
Step 1: If not already present, upload a PDB file to /storage/viet/FreeBindCraft/inputs/

 Upload PDB (0)

Use the **Upload PDB** control to select and upload the target PDB file from your local computer.



The uploaded file will be saved in the inputs/ directory within the FreeBindCraft project:



Step 2: Configure the Target Settings

The target settings JSON file specifies the target structure against which FreeBindCraft will generate binders. It also defines the binder naming prefix, output directory, target chains, hotspot residues, binder length range, and requested number of accepted designs.

If you do not already have a target settings JSON file, create one directly through the user interface by completing the following fields:

Field	Description
File Name	Target settings JSON filename.
design_path	Directory for outputs of a FreeBindCraft run.
binder_name	The prefix used for generated binder filenames, for example, IFIT5_FreeBindCraft_ewok-Binder.
starting_pdb	The uploaded target PDB file.
chains	The target chains to include, for example, A or A,C. Other chains will be ignored.
hotspots	The target residues intended for binder interaction, for example, A56,A60-65. Leave this field empty for automatic selection.
lengths	The minimum and maximum binder lengths, for example, [65,150].
Number designs	The requested number of accepted binder designs.

Or you can edit/create a new target JSON file.

How to fill in the target settings:

- **File Name** — Name for this settings file (e.g. `PDL1.json`). Start with a letter; use letters, numbers, `_`, or `-` only (max 150 characters). Don't include a folder path.
- **binder_name** — Short label used in output PDB names (e.g. `PDL1-Binder`). Start with a letter or number; no spaces or slashes (max 150 characters).
- **design_path** — Folder where results will be saved (e.g. `./outputs/PDL1_freebindcraft`). The last folder name follows the same rules as binder_name (path max 512 characters).
- **starting_pdb** — Pick your target structure from the files in `inputs/` (upload one above if the list is empty).
- **chains** — Which chain(s) to design against, separated by commas (e.g. `A` or `A,C`).
- **hotspots** — Optional. Leave blank to let the model choose a site, or specify residues such as `A56`, `A60-65`, or a whole chain `A`.
- **lengths** — Binder size range as min then max (e.g. `[65, 150]`). Min must be less than max; max cannot exceed 300.
- **num designs** — How many accepted designs to generate (1–100).

Problems are highlighted in red. Save Changes turns on once everything looks good.

File Name:	IFIT5_alphafoldmodel.json
design_path:	/storage/viet/FreeBindCraft/outputs/IFIT5_alphafoldmodel_freebindcraft
binder_name:	IFIT5_alphafoldmodel-Binder
starting_pdb:	IFIT5_alphafoldmodel.pdb
Refresh PDB list	
chains:	A
hotspots:	e.g. A56,A60-65 (empty = no preference)
lengths:	[65, 150]
num designs:	11
Save Changes	

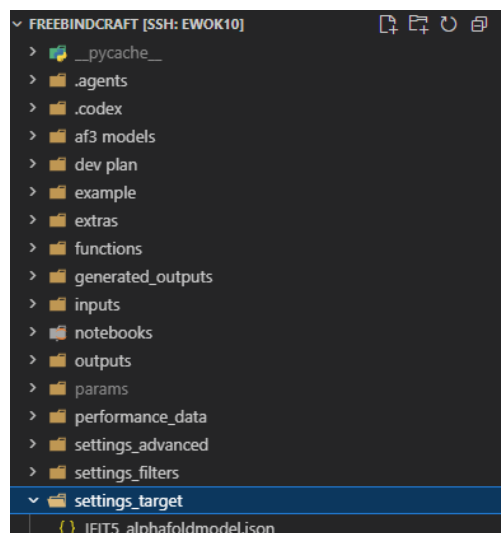
After completing the fields, click **Save Changes**.

Alternatively, upload an existing target settings JSON file that corresponds to the uploaded target PDB, rather than creating a new configuration manually.

Step 2: If not already present, upload a JSON file to `/storage/viet/FreeBindCraft/settings_target/`

Upload JSON (0)

Both uploaded and newly created target settings files are saved in the project's `settings_target/` directory.



Step 3-6: Run FreeBindCraft on specific target setting

Target setting selection

From the Target JSON dropdown menu, select the target settings file that you created or uploaded.

Select a target JSON file (refresh after upload/create):

Target JSON: IFIT5_cropped.json

Selected Target JSON: IFIT5_cropped.json

Refresh JSON

IFIT5_cropped_mem_usage.json

IFIT5_fl.json

PDL1.json

Configure the GPU

Select the GPU to use for the FreeBindCraft job.

FreeBindCraft options

GPU: GPU 0: NVIDIA GeForce RTX 4080 SUPER — 3688 MiB free / 16376 MiB

Refresh GPU list

Sets `CUDA_VISIBLE_DEVICES` for the FreeBindCraft job. Memory values are point-in-time snapshots; use Refresh GPU list to update them.

Review the **nvidia-smi** panel for information about the available GPUs and their current usage.

nvidia-smi details for the current selection

How to read **nvidia-smi**: The report lists every GPU visible on this machine. **Memory-Usage** is current VRAM use / capacity; **GPU-Util** shows how busy the GPU is: 0% means it is mostly idle, while 100% means it is very busy. This is separate from memory usage. **Pwr-Usage/Cap** is current power draw / power limit, and **Perf** is the performance state (P0 is highest performance). The **Processes** section shows programs currently using GPU memory.

Mon Sep 7 19:21:16 2026

NVIDIA-SMI 535.274.02 Driver Version: 535.274.02 CUDA Version: 12.2									
GPU	Name		Persistence-M	Bus-Id	Disp.A	Volatile	Uncorr.	ECC	
Fan	Temp	Perf	Pwr:Usage/Cap	Memory-Usage		GPU-Util	Compute M.		MIG M.

0	NVIDIA GeForce RTX 4080	P8	Off	00000000:0B:00.0	Off	0%	Default	N/A	
0%	46C		8W / 320W	1MiB / 16376MiB					

1	NVIDIA GeForce RTX 4080	P8	Off	00000000:41:00.0	Off	0%	Default	N/A	
0%	40C		11W / 320W	1MiB / 16376MiB					

Processes:									
GPU	GI	CI	PID	Type	Process name	GPU Memory Usage			
ID	ID								
=====									
No running processes found									

Binder generation against large target proteins can require substantial GPU memory. Enable **Track GPU memory usage** to monitor memory consumption and assess it against the selected GPU's capacity.

☒ Track GPU memory usage

Off by default. When enabled, BindCraft writes `gpu_memory_stats.csv` in the design folder.

The **Active FreeBindCraft jobs using GPU memory** panel displays FreeBindCraft jobs currently using GPU resources.

Active FreeBindCraft jobs using GPU memory

Active FreeBindCraft jobs: This table shows only FreeBindCraft processes currently using GPU memory. **GPU** is the GPU number, **PID** identifies the running process, **User** is the account running it, **Process** is the program name, **CWD** shows the project folder it is running from, and **GPU Memory** shows how much memory that job is using. If the table is empty, no FreeBindCraft job is currently using a GPU.

GPU	PID	USER	PROCESS	CWD	GPU MEMORY
0	22744	viet	python	/storage/viet/FreeBindCraft	12384 MiB

Click **Refresh GPU list** to update the GPU information and active-job displays.

FreeBindCraft options

GPU: GPU 0: NVIDIA GeForce RTX 4080 SUPER — 3688 MiB free / 16376 MiB

Refresh GPU list

Sets CUDA_VISIBLE_DEVICES for the FreeBindCraft job. Memory values are point-in-time snapshots; use Refresh GPU list to update them.

Output Artefact Settings

Rank by: i_pTM

Metric used to rank accepted final designs. i_pTM is the default; ipSAE ranks by interface predicted Structural Alignment Error.

☐ Verbose
☐ Disable plots
☐ Disable animations

Verbose: more detailed progress/timing logs. **Disable plots / animations:** skip saving trajectory plots and animations (faster, less disk use).

- Ranking metric:** Used to rank accepted final designs. Default to i_pTM
- Verbose:** Displays detailed progress and timing information in the logs.
- Disable plots:** Skips saving trajectory plots, reducing runtime and disk usage.
- Disable animations:** Skips saving trajectory animations.

Configure binder filtering and design validation

The **Filters** setting defines the quality criteria that binder designs must satisfy to be accepted. These criteria include confidence scores, interface quality, structural clashes, and binding energy.

The **Advanced** setting controls how FreeBindCraft generates and validates binders, including the design algorithm and the AlphaFold2 and ProteinMPNN settings.

Select a settings_filters file and a settings_advanced file, then generate the run script.

Filters: default_filters.json

Advanced: default_4stage_multimer.json

Generate BindCraft Run Script with Settings

For the default configuration, select default_filters.json under **Filters** and default_4stage_multimer.json under **Advanced**.

For further information, refer to the following resources:

- Filter settings:** <https://github.com/martinpacesa/BindCraft#filters>
- Advanced settings:** <https://github.com/martinpacesa/BindCraft#advanced-settings>
- BindCraft paper:** <https://doi.org/10.1101/2024.09.30.615802>

Generate the run script and launch the job

Once all settings have been configured, click **Generate FreeBindCraft Run Script with Settings**.

Generate FreeBindCraft Run Script with Settings

```
Generated command:

CUDA_VISIBLE_DEVICES=0
GPU memory tracking: disabled
python \
  -u \
  /storage/viet/FreeBindCraft/bindcraft.py \
  --settings \
```

Then click **Run FreeBindCraft** to launch the job using the generated script.

Press **Run FreeBindCraft** to start the job in a detached **tmux** session (survives notebook kernel death). Reattach with **tmux attach -t <job>**.

Run FreeBindCraft

```
Started tmux session: IFIT5_alphafoldmodel_freebindcraft
CUDA_VISIBLE_DEVICES=0
python -u /storage/viet/FreeBindCraft/bindcraft.py --settings /storage/viet/FreeBindCraft/settings_target/IFIT5_alphafoldmodel.json
Design path: /storage/viet/FreeBindCraft/outputs/IFIT5_alphafoldmodel_freebindcraft
Target accepted designs: 11
Log: /storage/viet/FreeBindCraft/outputs/IFIT5_alphafoldmodel_freebindcraft/freebindcraft_20260921_163044.log
Attach: tmux attach -t IFIT5_alphafoldmodel_freebindcraft
Kill: tmux kill-session -t IFIT5_alphafoldmodel_freebindcraft
```

Monitor and manage jobs

Monitor the selected job's progress in the status section at the bottom of the user interface.

Select a running job below to watch its **progress bar**, then click **Abort a running job** if you need to stop it. Use **Refresh Jobs Dropdown** if it is missing.

Select a job: IFIT5_cropped_bindcraft

Accepted:

IFIT5_cropped_bindcraft · 0/1 accepted designs · trajectories: 0 · current: **starting**

Abort a running job

Refresh Jobs Dropdown

Multiple FreeBindCraft jobs can run concurrently. Assign a different **design_path** to each job so that their outputs are stored in separate directories.

To stop a job, select it from the job dropdown menu and click **Abort a running job**. If a running job is not listed, click **Refresh Jobs Dropdown** to update the available selections.

Select a running job below to watch its **progress bar**, then click **Abort a running job** if you need to stop it. Use **Refresh Jobs Dropdown** if it is missing.

Select a job: IFIT5_cropped_250_freebindcraft

Accepted:

IFIT5_cropped_250_freebindcraft · 0/11 accepted designs · trajectories: 1 · current: **Stage 2: Softmax Optimisation**

Abort a running job

Refresh Jobs Dropdown

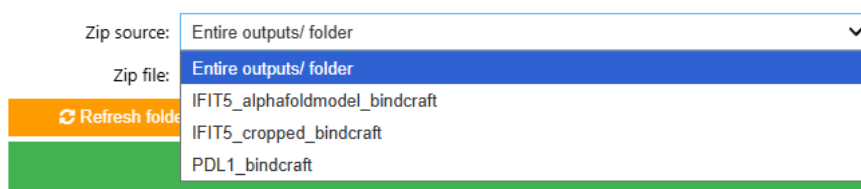
Aborted job 'IFIT5_alphafoldmodel_freebindcraft' (tmux session 'IFIT5_alphafoldmodel_freebindcraft').

Step 7: Compress and Download the Outputs

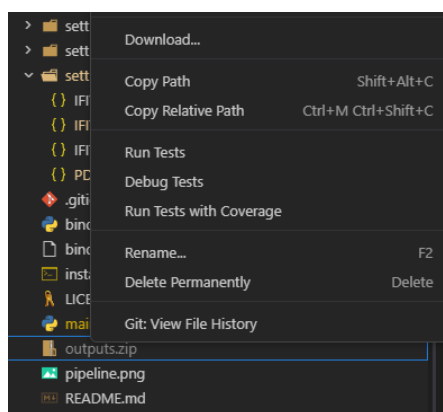
FreeBindCraft output files can be large. Compress the required outputs into a ZIP archive before downloading them.

In the **Zip source** dropdown menu, select either the entire outputs/ directory or an individual design output folder, then create the archive.

Choose the entire **outputs/** folder or a single design folder. The archive is written to the project root (parent of **notebooks/**): **/storage/viet/BindCraft**.



The ZIP archive will be saved in the FreeBindCraft project's root directory. To download it, locate the .zip file in the VS Code **Explorer** panel, right-click it, and select **Download....** Choose a destination on your local computer and save the file.



3.3 Licensing

Licensing

☒ `--no-pyrosetta`

FreeBindCraft is MIT-licensed (FreeBindCraft). Keep `--no-pyrosetta` checked for license-free work. PyRosetta is free for academic and non-profit use but requires a commercial license for non-academic use (see [PyRosetta licensing](#) and the [Rosetta Commons licensing FAQ](#)). If PyRosetta is needed, uncheck this option or use [BindCraft](#). For how PyRosetta steps are replaced with license-free alternatives, see the [technical overview](#).

Keep `--no-pyrosetta` **checked** for license-free use. If PyRosetta is required, uncheck `--no-pyrosetta` or use [BindCraft](#). For further information, see:

- **PyRosetta licensing:** <https://www.pyrosetta.org/home/licensing-pyrosetta>
- **Rosetta Commons licensing FAQ:** <https://rosettacommons.org/software/licensing-faq/>
- **FreeBindCraft PyRosetta bypass:** <https://github.com/cytokineking/FreeBindCraft#key-modification-pyrosetta-bypass-functionality>

- **Technical overview:**

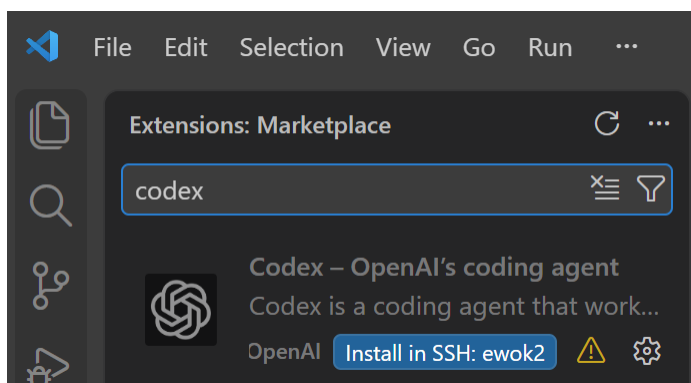
https://github.com/cytokineking/FreeBindCraft/blob/master/technical_overview/FreeBindCraft_Technical_Overview.md

4. Useful Extensions

Codex

If you have a ChatGPT account, you can optionally install the Codex extension on the Ewok host for AI-assisted work within the project.

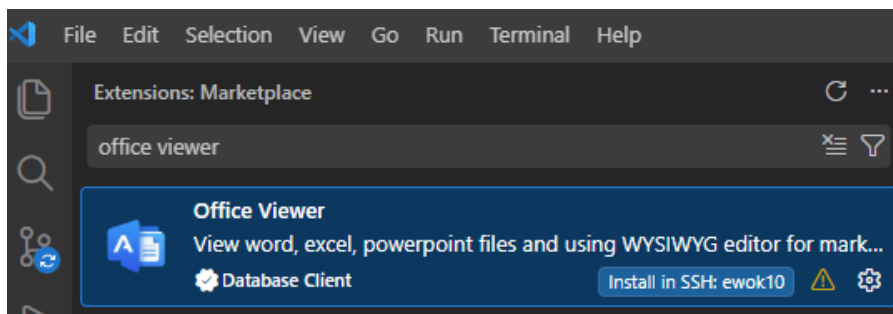
Extension: <https://marketplace.cursorapi.com/items/?itemName=openai.chatgpt>



Office Viewer

This extension helps with reading FreeBindCraft output csv files directly in VS Code.

Extension link: <https://marketplace.cursorapi.com/items/?itemName=cweijan.vscode-office>



final_design_stats.csv

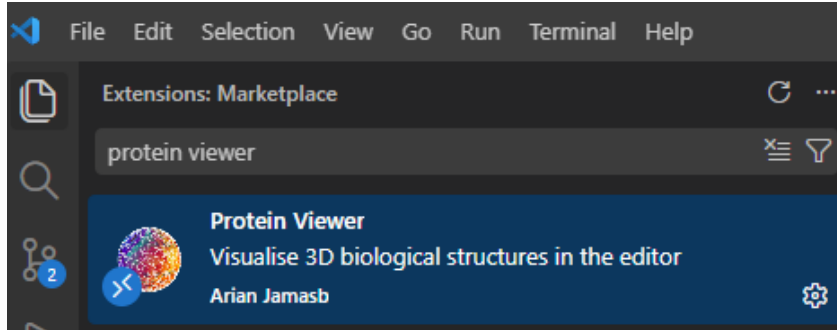
Rank

Rank	Design	Proto...	Length	Seed	Helic...	Target_Hots...	Sequence	InterfaceResid
1	IFITS_bindcraft_ewok-Binder_I145_s476...	4stage	145	4764...	-0.3		MKPDENISLEEAIKKLEELLEKMKKGFPKD...	B3,B33,B36,B3
2	IFITS_bindcraft_ewok-Binder_I93_s7941...	4stage	93	7941...	-0.3		MLDPSKYRTIEAWEKAYEELKKIAEEVLSDP...	B27,B33,B37,B
3	IFITS_bindcraft_ewok-Binder_I145_s476...	4stage	145	4764...	-0.3		MKPDENISLEEARKLLEELKQMENGEFPE...	B3,B33,B36,B3

Protein Viewer

This allows inspection of target protein or generated binders PDB directly in VS Code.

Extension link: <https://marketplace.visualstudio.com/items?itemName=ArianJamasp.protein-viewer>



Right click on a PDB and select *Launch Protein Viewer* to view its 3D structure.

